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①

$$\text{Var}(b_1|X) = \frac{\sigma^2}{\sum (X_i - \bar{X})^2}$$

$$\sum_{t=1}^T (t - \bar{t})^2 = \sum_{t=1}^T t^2 - \frac{(\sum_{t=1}^T t)^2}{T}$$

$$\text{其中 } \sum_{t=1}^T t^2 = \frac{T(T+1)(2T+1)}{6}, \quad \sum_{t=1}^T t = \frac{T(T+1)}{2}$$

$$\Rightarrow \sum_{t=1}^T (t - \bar{t})^2 = \frac{T(T+1)(2T+1)}{6} - \frac{T^2(T+1)^2}{4T} = \frac{T(T+1)}{2} \left[\frac{2T+1}{3} - \frac{(T+1)}{2} \right] = \frac{T(T+1)(T-1)}{12} = \frac{T(T^2-1)}{12}$$

$$\Rightarrow \text{Var}(b_1|X) = \frac{\sigma^2}{\frac{T(T^2-1)}{12}} = \frac{12\sigma^2}{T(T^2-1)}$$

$\Rightarrow \lim_{T \rightarrow \infty} \text{Var}(b_1|X) = 0$ ，估計量滿足一致性 #

②

$$\sum_{t=1}^T X_t = 0.5 + 0.5^2 + 0.5^3 + \dots + 0.5^T = \frac{0.5(1-0.5^T)}{1-0.5} = 1-0.5^T$$

$$\sum_{t=1}^T (X_t)^2 = \sum_{t=1}^T (0.25)^t = \frac{0.25(1-0.25^T)}{1-0.25} = \frac{1}{3}(1-0.25^T)$$

$$\sum_{t=1}^T (X_t - \bar{X})^2 = \sum_{t=1}^T (X_t)^2 - \frac{(\sum_{t=1}^T X_t)^2}{T} = \frac{1}{3}(1-0.25^T) - \frac{(1-0.5^T)^2}{T}$$

$$\text{Var}(b_1|X) = \frac{\sigma^2}{\frac{1}{3}(1-0.25^T) - \frac{(1-0.5^T)^2}{T}}$$

$\lim_{T \rightarrow \infty} \text{Var}(b_1|X) = \frac{\sigma^2}{\frac{1}{3}} \neq 0$ ，估計量不具一致性 #

③

模型①的隨機趨勢項隨樣本增加而發散，提供了無限資訊來鑑定斜率；而模型②的幾何級數項快速衰減，導致新增的樣本無法提供有效的新資訊，因此無法消除估計誤差，不具備一致性。

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$$\textcircled{1} \begin{cases} H_0: \beta_2 = 0 \\ H_a: \beta_2 \neq 0 \end{cases}$$

$$2. \alpha = 0.05$$

$$3. \text{test statistic: } t = \frac{b_2 - 0}{\text{se}(b_2)} \stackrel{H_0}{\sim} t(60)$$

$$4. t^* = \frac{3}{2} = 1.5$$

$$5. RR = \{t \mid |t| \geq t_{0.025}(60) = 2\}$$

6. $t^* \notin RR$, do not reject H_0 . β_2 並未顯著異於 0 吖

$$\textcircled{2} \begin{cases} H_0: \beta_1 + 2\beta_2 = 5 \\ H_a: \beta_1 + 2\beta_2 \neq 5 \end{cases}$$

$$2. \alpha = 0.05$$

$$3. \text{test statistic: } t = \frac{b_1 + 2b_2 - 5}{\text{se}(b_1 + 2b_2)} \stackrel{H_0}{\sim} t(60) \quad \text{其中 } \text{se}(b_1 + 2b_2) = \sqrt{\widehat{\text{Var}}(b_1) + 4\widehat{\text{Var}}(b_2) + 4\widehat{\text{Cov}}(b_1, b_2)}$$

$$= \sqrt{3 + 16 + 8} \\ = \sqrt{11}$$

$$4. t^* = \frac{3}{\sqrt{11}} \approx 0.9045$$

$$5. RR = \{t \mid |t| \geq t_{0.025}(60) = 2\}$$

6. $t^* \notin RR$, do not reject H_0 . $\beta_1 + 2\beta_2$ 並未顯著異於 5 吖

$$\textcircled{3} \begin{cases} H_0: \beta_1 - \beta_2 + \beta_3 = 4 \\ H_a: \beta_1 - \beta_2 + \beta_3 \neq 4 \end{cases}$$

$$2. \alpha = 0.05$$

$$3. t_{\text{stat}} = \frac{b_1 - b_2 + b_3 - 4}{\text{se}(b_1 - b_2 + b_3)} \stackrel{H_0}{\sim} t(60)$$

$$\text{其中 } \text{se}(b_1 - b_2 + b_3) = \sqrt{\widehat{\text{Var}}(b_1) + \widehat{\text{Var}}(b_2) + \widehat{\text{Var}}(b_3) - 2\widehat{\text{Cov}}(b_1, b_2) - 2\widehat{\text{Cov}}(b_1, b_3) + 2\widehat{\text{Cov}}(b_2, b_3)}$$

$$= \sqrt{16} = 4$$

$$4. t^* = \frac{-2 - 4}{4} = -1.5$$

$$5. RR = \{t \mid |t| \geq t_{0.025}(60) = 2\}$$

6. $t^* \notin RR$, do not reject H_0 . $\beta_1 - \beta_2 + \beta_3$ 並未顯著異於 4 吖

(5.20)

#a~d 小題

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$$\textcircled{a} \quad \bar{y}_1 = B_1 + B_2 \bar{x}_1 + \bar{e}_1$$

$$\bar{y}_2 = B_1 + B_2 \bar{x}_2 + \bar{e}_2$$

$$\hat{B}_{2, \text{mean}} = \frac{(B_1 + B_2 \bar{x}_2 + \bar{e}_2) - (B_1 + B_2 \bar{x}_1 + \bar{e}_1)}{\bar{x}_2 - \bar{x}_1} = B_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}$$

$$E(\hat{B}_{2, \text{mean}} | X) = B_2 \quad \# \quad \text{其中 } E(e_i | X) = 0$$

Therefore, $\hat{B}_{2, \text{mean}}$ is an unbiased estimator of B_2

$$\textcircled{b} \quad \text{Var}(\hat{B}_{2, \text{mean}} | X) = \text{Var}\left(B_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}\right)$$

$$= \frac{\left(\frac{2\sigma^2}{N} + \frac{2\sigma^2}{N}\right)}{(\bar{x}_2 - \bar{x}_1)^2}$$

$$= \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2} \quad \#$$

$$\text{其中 } \text{Var}(\bar{e}_1) = \text{Var}\left(\frac{1}{N/2} \sum_{i=1}^{N/2} e_{i,1}\right) = \frac{\sum_{i=1}^{N/2} \text{Var}(e_{i,1})}{(N/2)^2} = \frac{(N/2)\sigma^2}{(N/2)^2} = \frac{2\sigma^2}{N}$$

$$\text{又 } \text{Var}(\bar{e}_1) = \text{Var}(\bar{e}_2) = \frac{2\sigma^2}{N}$$

③ Law of Total Variance:

$$\text{Var}(\hat{B}_{2, \text{mean}}) = E[\text{Var}(\hat{B}_{2, \text{mean}} | X)] + \text{Var}[E(\hat{B}_{2, \text{mean}} | X)] = E\left(\frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}\right) + \underbrace{\text{Var}(B_2)}_0 = E\left[\frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}\right] \quad \#$$

④

已知 $E(\hat{B}_{2, \text{mean}} | X) = B_2$ 滿足不偏性

當 N 趨近於 ∞ ($N \rightarrow \infty$), $\text{Var}(\hat{B}_{2, \text{mean}})$ 趨近於 0

$$\lim_{N \rightarrow \infty} \text{Var}(\hat{B}_{2, \text{mean}}) = 0$$

$\Rightarrow$ 當 $N \rightarrow \infty$ 時, 只要自變數 X 的分佈能確保 $N(\bar{x}_2 - \bar{x}_1)^2 \rightarrow \infty$, 則 $\hat{B}_{2, \text{mean}}$ 就是一致估計量。#

#e

```
>
> # e輸出結果
> print(results)
      N var_b2_OLS var_beta2_mean estimate_inv_sx2 estimate_mean_diff_part
1  100 1.24362517      1.6631963      0.1243625      0.1663196
2   500 0.24402399      0.3260404      0.1220120      0.1630202
3  1000 0.12290628      0.1656282      0.1229063      0.1656282
4  5000 0.02413174      0.0323161      0.1206587      0.1615805
>
> # var_b2_OLS 對應var(b2 | x)
> # var_beta2_mean 對應var(beta2_mean | x)
> # estimate_inv_sx2 對應(sx^2)^-1 的估計
> # estimate_mean_diff_part 對應4 / (x2_bar - x1_bar)^2 的估計
>
```

#f

根據模擬數據，首先在 i 部分，OLS 估計量的變異數 $\text{var}(b_2 | x)$ 在所有樣本數下均小於 $\text{var}(b_2, \text{mean_hat} | x)$ ，這說明了 OLS 比均值估計量更具效率。在 ii 部分， $\{(s_x)^2\}^{(-1)}$ 的數值穩定在 0.12 左右，而 $4/(x_{\text{bar}_2} - x_{\text{bar}_1})^2$ 則穩定在 0.16 左右；這顯示 $b_2, \text{mean_hat}$ 的變異數（由 0.16 組成）本生就比 OLS（由 0.12 組成）來得大。

隨著樣本數 N 的增加，i 部分的兩個變異數皆顯著遞減並趨近於 0，而 ii 部分的兩個估計值則保持穩定。這意味著變異數的下降主要是由分母中的 N 增加所驅動的。由於當 N 趨近於無限大時，變異數會趨近於零，因此可以斷定 $b_2, \text{mean_hat}$ 是一個一致估計量。

#g #h

(g)

$$X \sim \mathcal{N}(0, 10)$$
$$\text{Var}(X) = \frac{(10-0)^2}{12} = \frac{100}{12}$$
$$S_x^2 \xrightarrow{P} \frac{100}{12}$$

根據 Continuous Mapping Theorem

$$E\left(\frac{1}{S_x^2}\right) \xrightarrow{P} \frac{1}{\left(\frac{100}{12}\right)} = 0.12 \#$$

$$\begin{cases} \bar{X}_1 = 2.5 \\ \bar{X}_2 = 7.5 \end{cases} \Rightarrow \bar{X}_2 - \bar{X}_1 \xrightarrow{P} (7.5 - 2.5) = 5$$
$$E\left(\frac{4}{(\bar{X}_2 - \bar{X}_1)^2}\right) \xrightarrow{P} \frac{4}{5^2} = \frac{4}{25} = 0.16 \#$$

(h)

若不按大小排序而隨機分組，則 $\begin{cases} X_1 \xrightarrow{P} E(X) = 5 \\ X_2 \xrightarrow{P} E(X) = 5 \end{cases}$

導致分母的平方項 $(\bar{X}_2 - \bar{X}_1)^2 \xrightarrow{P} (5 - 5)^2 = 0$

由於分母趨近於 0，根據連續映射定理，該估計量的變異數會趨向 ∞ ，因此不具一致性。#